MAT092: Remedial Course in Mathematics

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LECTURE-04

Motivation

Functions describe how elements of one set (domain) correspond to another (codomain). But often, the **cardinality** of these sets already hints at whether the function is:

injective, surjective, or bijective.

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Key Idea

Comparing $|A|$ and $|B|$ gives valuable information about the possible type of function.

Recap: Function Properties

Property	**Definition**	**Symbolic Form**
Injective (One-to-one) give different outputs	Different inputs $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$	
Surjective (Onto) of codomain is hit	Every element $\forall y \in B,\ \exists x \in A : f(x) = y$	
Bijective and surjective	Both injective One-to-one and onto	

Cardinality Clues

Let $f : A \to B$, with $|A| = m$ and $|B| = n$.

Type	**Relation**	**Consequence**
Injective	$m \leq n$	Domain can't have more elements than codomain.
Surjective	$m \geq n$	Domain must be large enough to cover all of codom
Bijective	$m = n$	Perfect pairing between domain and codomain.

Examples

Example 1

$f : \{1, 2, 3\} \to \{a, b, c, d\}$
$|A| = 3, |B| = 4 \Rightarrow$ can be injective, not surjective.

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$|A| = 4, |B| = 3 \Rightarrow$ can be surjective, not injective.

Example 3

$f : \{1, 2, 3\} \to \{a, b, c\}$
$|A| = |B| \Rightarrow$ possible bijection.

Infinite Sets: A Remark

Finite vs Infinite

For finite sets, counting elements works.

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Example

There exists a bijection between $\mathbb{N}$ and $2\mathbb{N}$:

$$f(n) = 2n.$$

Even though $2\mathbb{N} \subset \mathbb{N}$, they have the same cardinality.

Summary Table

Property	**Finite Sets**	**Infinite Sets**
Injective	$\lvert A\rvert \leq \lvert B\rvert$	Exists $A \hookrightarrow B$
Surjective	$\lvert A\rvert \geq \lvert B\rvert$	Exists $A \twoheadrightarrow B$
Bijective	$\lvert A\rvert = \lvert B\rvert$	Exists $A \leftrightarrow B$

Quick Quiz

$f : \{1, 2, 3, 4, 5\} \to \{a, b, c\}$ Can it be injective? No.

Can it be surjective? Yes.

$f : \{1, 2\} \to \{a, b, c\}$ Can it be injective? Yes. Surjective? No.

Quick Quiz

$f : \{1, 2, 3, 4, 5\} \to \{a, b, c\}$ Can it be injective? No.

Can it be surjective? Yes.

$f : \{1, 2\} \to \{a, b, c\}$ Can it be injective? Yes. Surjective? No.

Takeaway

Cardinality gives strong hints—but the mapping itself confirms the property!

Composition Function

$f = \{(1,1),(2,3),(3,1),(4,2)\}$, and $g = \{(1,2),(2,3),(3,1),(4,2)\}$, then $g \circ f = \{(1,2),(2,1),(3,2),(4,3)\}$

Quadratic Functions: A Classic Example

Definition: A quadratic function is a polynomial of degree 2:

$$f(x) = ax^2 + bx + c, \quad a \neq 0.$$

Graph: Its graph is a parabola:

Opens upwards if $a > 0$, downwards if $a < 0$.

Axis of symmetry: $x = -\frac{b}{2a}$.

Vertex: $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$.

Example:

$$f(x) = x^2 - 4x + 3 \Rightarrow a = 1, b = -4, c = 3.$$

Vertex at $(2, -1)$, roots at $x = 1, 3$.

Quadratic Equations and Their Roots

Equation:

$$ax^2 + bx + c = 0, \quad a \neq 0.$$

Solutions (roots) given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Discriminant: $\Delta = b^2 - 4ac$

- $\Delta > 0$: Two distinct real roots.
- $\Delta = 0$: One repeated real root.
- $\Delta < 0$: Two complex conjugate roots.

Example:

$$x^2 - 4x + 3 = 0 \Rightarrow \Delta = 4.$$

Hence $x = 1, 3$.

Remark: Quadratic functions encode symmetry — the vertex sits halfway between the roots.

Completing the Square: Geometry in Algebra

Idea: Rewrite $ax^2 + bx + c$ to reveal symmetry.

$$f(x) = a\left(x^2 + \frac{b}{a}x + \frac{c}{a}\right) = a\left(\left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a^2}\right).$$

Hence:

$$f(x) = a(x-h)^2 + k, \quad \text{where } h = -\frac{b}{2a},\ k = f(h).$$

Geometric meaning:

(h, k) = vertex of the parabola.

The parabola is symmetric about $x = h$.

Example:

$$f(x) = x^2 - 4x + 3 = (x-2)^2 - 1.$$

Vertex $(2, -1)$, minimum value -1.

Why Quadratics Matter

Intuition: Quadratics are everywhere because they describe:

Equilibrium and optimization: minimizing cost, energy, or distance.

Motion under uniform acceleration: $s(t) = ut + \frac{1}{2}at^2$.

Geometry: circles, conics, and symmetry.

Mathematical beauty: Quadratic functions are the simplest nonlinear functions, yet rich enough to capture curvature and symmetry.

Moral: Quadratic functions bridge algebra, geometry, and physics — a perfect meeting point of structure and intuition.